In [1]:
```
import pandas as pd
```

In [2]:
```
import numpy as np
```

In [3]:
```
import matplotlib.pyplot as plt
```

In [6]:
```
data=pd.read_csv(r"C:\Users\praka\Downloads\Dataset .csv")
```

In [7]:
```
data.head()
```

Out[7]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longituc
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.0275
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.0141
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.0568
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.0564
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.0575

5 rows × 21 columns

In [8]:
```
data.tail()
```

Out[8]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Loc Verb
9546	5915730	Naml\ Gurme	208	––stanbul	Kemanke–– Karamustafa Pa––a Mahallesi, R\ht\m ...	Karak–_y	Karak• ––sta
9547	5908749	Ceviz A––ac\	208	––stanbul	Ko––uyolu Mahallesi, Muhittin ––st–_nda–– Cadd...	Ko––uyolu	Ko––u\ ––sta
9548	5915807	Huqqa	208	––stanbul	Kuru–_e––me Mahallesi, Muallim Naci Caddesi, N...	Kuru–_e––me	Kuru–_e––– ––sta
9549	5916112	A–––k Kahve	208	––stanbul	Kuru–_e––me Mahallesi, Muallim Naci Caddesi, N...	Kuru–_e––me	Kuru–_e––– ––sta
9550	5927402	Walter's Coffee Roastery	208	––stanbul	Cafea––a Mahallesi, Bademalt\ Sokak, No 21/B, ...	Moda	M ––sta

5 rows × 21 columns

In [9]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
Restaurant ID           9551 non-null int64
Restaurant Name         9551 non-null object
Country Code            9551 non-null int64
City                    9551 non-null object
Address                 9551 non-null object
Locality                9551 non-null object
Locality Verbose        9551 non-null object
Longitude               9551 non-null float64
Latitude                9551 non-null float64
Cuisines                9542 non-null object
Average Cost for two    9551 non-null int64
Currency                9551 non-null object
Has Table booking       9551 non-null object
Has Online delivery     9551 non-null object
Is delivering now       9551 non-null object
Switch to order menu    9551 non-null object
Price range             9551 non-null int64
Aggregate rating        9551 non-null float64
Rating color            9551 non-null object
Rating text             9551 non-null object
Votes                   9551 non-null int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

In [10]:
```
data.shape
```

Out[10]:
```
(9551, 21)
```

In [11]:
```
data.describe()
```

Out[11]:

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	Aggre ra
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	9551.00
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	2.66
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	1.51
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	0.00
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	2.50
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	3.20
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	3.70
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	4.90

In [12]:
```
data.isnull()
```

Out[12]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	C
0	False	False	False	False	False	False	False	False	False	
1	False	False	False	False	False	False	False	False	False	
2	False	False	False	False	False	False	False	False	False	
3	False	False	False	False	False	False	False	False	False	
4	False	False	False	False	False	False	False	False	False	
...	...	...	...	...	...	...	...	...	...	
9546	False	False	False	False	False	False	False	False	False	
9547	False	False	False	False	False	False	False	False	False	
9548	False	False	False	False	False	False	False	False	False	
9549	False	False	False	False	False	False	False	False	False	
9550	False	False	False	False	False	False	False	False	False	

9551 rows × 21 columns

In [14]:
```
data.isnull().sum()
```

Out[14]:
```
Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Is delivering now       0
Switch to order menu    0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

In [15]:
```
data.duplicated()
```

Out[15]:
```
0       False
1       False
2       False
3       False
4       False
        ...
9546    False
9547    False
9548    False
9549    False
9550    False
Length: 9551, dtype: bool
```

In [17]:
```
data.duplicated().sum()
```

Out[17]:
```
0
```

In [23]:
```
data.dtypes
```

Out[23]:
```
Restaurant ID             int64
Restaurant Name          object
Country Code              int64
City                     object
Address                  object
Locality                 object
Locality Verbose         object
Longitude               float64
Latitude                float64
Cuisines                 object
Average Cost for two      int64
Currency                 object
Has Table booking        object
Has Online delivery      object
Is delivering now        object
Switch to order menu     object
Price range               int64
Aggregate rating        float64
Rating color             object
Rating text              object
Votes                     int64
dtype: object
```

In [24]:
```
#data type conversion
```

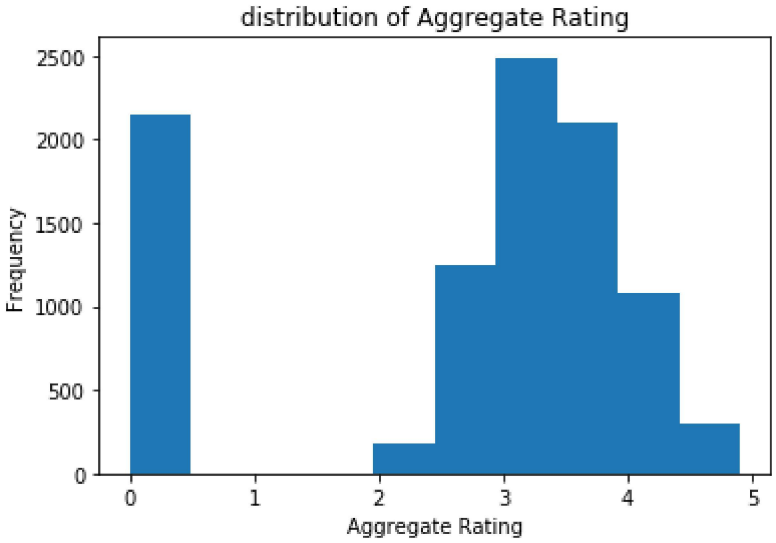
In [25]:
```
data['Aggregate rating']=data['Aggregate rating'].astype(float)
```

In [26]:
```
if 'City' in data.columns:data['City'].astype('category')
```

In [27]:
```
if 'Country Code' in data.columns:data['Country Code']=data['Country Code'].as
type('category')
```

In [28]:
```
#distribution of target variables
```

In [30]:
```
plt.hist(data['Aggregate rating'],bins=10)
plt.xlabel("Aggregate Rating")
plt.ylabel("Frequency")
plt.title("distribution of Aggregate Rating")
plt.show()
```



In [31]:
```
#class imbalance check
```

In [35]:
```
data['Aggregate rating'].value_counts()
```

Out[35]:
```
0.0    2148
3.2     522
3.1     519
3.4     498
3.3     483
3.5     480
3.0     468
3.6     458
3.7     427
3.8     400
2.9     381
3.9     335
2.8     315
4.1     274
4.0     266
2.7     250
4.2     221
2.6     191
4.3     174
4.4     144
2.5     110
4.5      95
2.4      87
4.6      78
4.9      61
2.3      47
4.7      42
2.2      27
4.8      25
2.1      15
2.0       7
1.9       2
1.8       1
Name: Aggregate rating, dtype: int64
```

In [ ]: